PNAS RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES OPEN ACCESS

Self-views converge during enjoyable conversations

Christopher Welker[a,1], Thalia Wheatley[a,b], Grace Cason[a], Catherine Gorman[a], and Meghan Meyer[c]

Edited by Susan Fiske, Princeton University, Jamaica, VT; received January 9, 2024; accepted September 4, 2024

Based on current research, it is evident that the way people see themselves is shaped by their conversation partners. Historically, this literature focuses on how one individual's expectations can shape another person's self-views. Given the reciprocal nature of conversation, we wondered whether conversation partners' self-views may mutually evolve. Using four-person round-robin conversation networks, we found that participants tended to have more similar self-views post-conversation than pre-conversation, an effect we term "inter-self alignment." Further, the more two partners' self-views aligned, the more they enjoyed their conversation and were inclined to interact again. This effect depended on both conversation partners becoming aligned. These findings suggest that the way we see ourselves is coauthored in the act of dialogue and that as shared self-views develop, the desire to continue the conversation increases.

conversation | self | social cognition | self-perception | homophily

Significance

Individuals tend to connect with those who are like them. This tendency is so strong that it is an organizational principle in social networks, termed homophily. Although past research has framed homophily as a selection-based process (i.e., searching for similar others), we asked whether people's self-views may also spontaneously become more similar through conversation. We find that conversation partners make more similar trait ratings of themselves after conversing than before and that this process also promotes bonding: The more similar partners' self-views become during conversation, the more they enjoy their time together and want to meet again. Our findings suggest that homophilic friendships may not be wholly deterministic. Instead, through conversation, we can become the company we keep.

The people we interact with shape who we are. Although some aspects of our self-views are stable over time (1–4), self-views can shift in meaningful social interactions and different self-dimensions are activated in different social contexts (5–9). Despite extensive research on this topic [for review, see (10)], it is not known whether people mutually adapt their self-views to one another during interaction (11). These joint influences may emerge in conversation—the primary way humans interact to maintain social bonds (12–14)—as it is characterized by a dynamic exchange between people. Given the reciprocal nature of conversation, do people's views of themselves coevolve? Here, we test whether people's self-views converge during conversation.

Conversations are known to align many behavioral and psychological processes from how people speak (15–18) to how they understand the world (19–22). The possibility that self-views also converge during conversations is consistent with research on how people change during friendship. Although people tend to befriend similar others (23–27), it is also true that friends influence each other to behave similarly (28, 29) and even come to view themselves as more similar over time (30). For example, friends' self-reported extraversion tends to become more similar over months and years of friendship (30). While prior work has demonstrated that established relationships can induce similarity, we investigate whether this can happen during a single conversation.

The possibility that conversations align self-views is also consistent with research on social influence (31–33). As early as 1936, Sherif demonstrated that people turn to others for guidance on how to perceive ambiguous visual stimuli, converging their perceptions through conversation (34). More recent work shows that dyadic conversations can homogenize beliefs across a social network (35). This suggests that conversations may generate self-other similarity at both the dyadic level (between conversation partners) and group level (between the self and group members more generally). If so, this could help us understand the connection between two known influences on social ties: 1) proximity—people tend to form ties with whom they are physically close (36–39), and 2) homophily—people who are similar to one another cluster together in social networks (23–26). Conversations between proximal community members may bring self-views into alignment at both the dyadic and community level, which over time may create homophily in social networks.

Consistent with these ideas, shared reality theory posits that people are intrinsically motivated to find common ground and to perceive that their own thoughts and inner experiences are aligned with others (40–42). Indeed, shared experience activates the brain's reward system supporting positive affect (43), which may reflect the intrinsic value of shared reality. Given that people tend to feel happier after conversations than before (44), we asked not only whether conversation aligns self-views, but whether this alignment predicts conversational enjoyment.

Of course, conversation is not monolithic (45). Conversations vary on many dimensions such as the goals of the conversation partners, the number of people involved, and the culture in which they take place (46–48). In particular, recent work has shown that conversations on "deep" topics that involve self-disclosure (e.g., sharing an embarrassing memory) lead to

Author affiliations: [a]Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH 03755; [b]Santa Fe Institute, Santa Fe, NM 87501; and [c]Department of Psychology, Columbia University, New York, NY 10027

Author contributions: C.W., T.W., and M.M. designed research; C.W., G.C., and C.G. performed research; C.W. analyzed data; and C.W., T.W., and M.M. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

[1]To whom correspondence may be addressed. Email: christopher.ll.welker@gmail.com.

This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2321652121/-/DCSupplemental.

Published October 14, 2024.

increases in conversation partners' perceived closeness relative to "shallow" conversations that are more impersonal (49–51). Recent work has also shown that self-views are most likely to change when participants are actively aware that they are reflecting on themselves (52). We posit that self-disclosure and self-reflection may be more likely to take place during deep (relative to shallow) conversations. We manipulated topic depth with the prediction that deeper conversations would involve greater convergence of self-views and would be more enjoyable.

Here, we examine whether conversation partners' self-views become more similar over the course of a conversation—a process we call "inter-self alignment." Moreover, we hypothesize that greater inter-self alignment predicts more enjoyable conversations and that deeper conversations evoke greater inter-self alignment and a stronger relationship between inter-self alignment and enjoyment. To test these possibilities, participants completed a series of trait ratings before and after having 10-min conversations about either shallow or deep topics. Participants also rated how enjoyable they found the conversations. We then quantified the distance between partners' trait ratings before and after each conversation and determined whether those distances increased (inter-self divergence) or decreased (inter-self alignment) over the course of the conversation. We found evidence of inter-self alignment and that conversations with greater inter-self alignment were rated as more enjoyable. These effects were equally robust in deep and shallow conversations, suggesting that even casual discussions may bring self-views into alignment. Further analyses explored whether 1) alignment was predictive of enjoyment at the level of the dyad or the individual, 2) participants' ability to accurately predict their partners' self-views mediated the relationship between inter-self alignment and enjoyment, 3) specific dimensions of the self were impacted more by alignment, and 4) alignment between conversation dyads normalized self-views across the sample.

Methods

Participants. We recruited 104 participants (60.4% White/Caucasian, 4.2% Black/African American, 22.9% Asian, and 12.5% Other/Mixed Race and 67.7% female with a mean age of 19.22 years) via an online participant recruitment system at Dartmouth. All participants were at least 18 years old and were compensated with course credit. Data collection and experimental procedures were approved by the Committee for the Protection of Human Subjects at Dartmouth in Hanover, New Hampshire (CPHS ID: 00030517), and all participants provided informed consent before taking part in the study.

The sample size was determined based on results from a pilot study and is recorded in our preregistration (AsPredicted #103006). Specifically, a separate set of 40 participants completed the conversation prompts and self-report scales of conversation enjoyment, closeness, and the trait ratings described below (*SI Appendix*, Supplementary Materials for a full description of the pilot study). Using the linear mixed model described in our results section below and reported on in our preregistration, we observed a medium-sized relationship between inter-self alignment and conversation enjoyment in the pilot sample [$\beta = 0.30$, 95% CI = (0.00,0.60), $P = 0.053$]. We therefore ran power analyses using the R package simr to establish an appropriate sample size for Study 1 based on effect sizes from the pilot study (Green & MacLeod, (53)). We found the sample size needed to achieve 80% power to detect a difference between our main model with and without inter-self alignment as a predictor, testing whether including inter-self alignment significantly improved our ability to predict enjoyment. We then added one more pair of groups (one for each condition) to account for the fact that significant effect sizes from pilot studies are often overestimates of the true effect size. Based on our power analyses, we determined that we needed a sample size of 24 groups (144 conversations, 96 participants) to detect robust results.

Participants were recruited in groups of four and had conversations in a round-robin structure [i.e., each member of the group had a one-on-one conversation with every other group member (Fig. 1)]. If a group either did not finish all their conversations or was missing at least two survey responses, it was excluded from further analyses. Two groups failed to meet the criteria for inclusion, leaving us with a final sample size of 96 participants (24 groups; 144 one-on-one conversations).

The majority of our participants had not interacted with each other prior to their conversations. When participants rated how well they knew their partner before taking part in the study on a scale from 0 (Not at all) to 100 (Extremely) with a midpoint at 50 (Somewhat), the average response was an 8.29 (SD = 21.35) and over half of the participants (55.21%) responded with 0. These numbers are on par with other work studying conversations between strangers (54). However, we control for the strength of any prior relationships between conversation partners by including the responses to this question as a covariate in our regression models.

Procedure. Each participant was assigned to a group with three other participants and, over the course of the study, had a 10-min conversation with every other member of their group. We employed a between-subjects design; four-person groups were randomly assigned to talk about either deep or shallow topics during their conversations. The conversation prompts were selected from those used in a previous study of deep and shallow conversations (55). Other than providing these prompts, participants were allowed to converse freely.

The study consisted of three phases (Fig. 1, *Right*). In the first phase, participants were sent a baseline survey with a 60-item trait scale (56) as well as the following personality trait scales: the self-monitoring scale (57), the MacArthur Scale of Subjective Social Status (58), the Revised UCLA Loneliness Scale (59), and a self-report of social variety seeking (60). We do not report results here from the personality trait scales as they were addressing a separate theoretical question. Participants also answered several demographic questions (age, race, gender, education). Participants were sent the survey at least 24 h before their first conversation and finished the survey prior to that conversation.

In the second phase, participants had three conversations–one with each other member of their group–with each conversation occurring on separate days. Although the conversations in our pilot study took place in person, the full

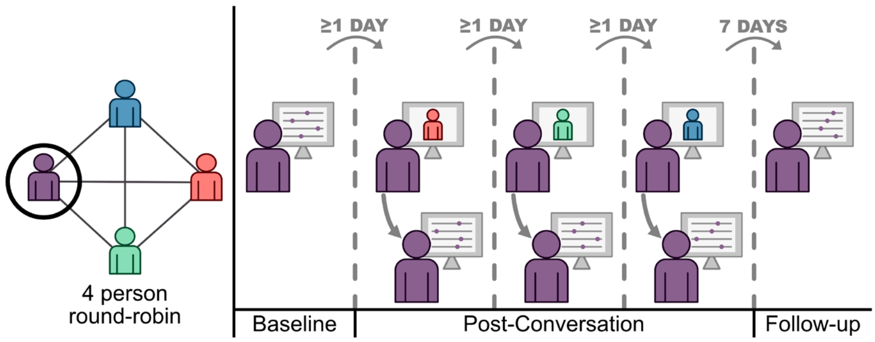

Fig. 1. (*Left*) Participants were recruited in groups of four and had conversations in a round-robin structure. Each participant had a one-on-one conversation with every other member of their group. (*Right*) Depiction of an example participant's experience in the experiment. Participants were sent the baseline survey by email. At least one day later, they participated in a dyadic conversation with one of their group members, after which they filled out the post-conversation survey. They repeated this step two more times, once with each remaining group member. All conversations took place on separate days. Finally, one week after their final conversation, participants were sent the follow-up survey.

round-robin paradigm was collected via a virtual video chat platform (Zoom) since it occurred during the COVID-19 pandemic. Before each conversation, participants were instructed to hide "self view" and expand the video window to fill the whole screen so that the experience would better resemble a face-to-face conversation. That is, they saw their conversation partner, but not themselves, during the conversation. The experimenter then briefly introduced the task, sent the participants the conversation prompts via the text chat, and left the virtual room. After 10 min, the experimenter rejoined the room and sent the participants a link to a survey in which they filled out the 60-item trait scale another time along with questions about how many of the prompts they discussed during their conversation, the depth of their conversation, how well they knew their partner before the study, how much they enjoyed the conversation, how close they felt to their partner, and how similar they felt to their partner. In addition, participants rated their partner using the same 60-item trait scale. For additional information about the exact wording and range of the survey questions, please see our OSF repository (https://osf.io/7wtz2/?view_only=8ee86d31444649d9a9a9ed218b932883).

In the third phase, the 60-item trait scale was sent to participants 7 d after their final conversation. Importantly, the display order of the items in the 60-item trait scale was randomized every time it was presented to counter any effects of taking the survey multiple times.

Variable Creation. All variables were created in R (version 4.2.2).

Inter-self alignment. We defined inter-self alignment as the degree to which conversation partners' self trait ratings became more similar after the conversation compared to at baseline. Following steps detailed in our preregistration, we operationalized inter-self alignment by comparing the Manhattan distance (Fig. 2) between the partners' self-reported trait ratings in the baseline surveys to the distance between their self-reported trait ratings in the post-conversation surveys. Our use of Manhattan distance was inspired by prior work using similar scales (56, 61); however, our results are not dependent on the use of this specific distance metric (*SI Appendix*, Supplementary Materials for additional analyses). We subtracted the distance between pairs' self ratings *after* their conversation from the distance between their self ratings *before* the conversation. This subtraction resulted in a value which, when positive, indicates that the pairs' trait ratings were more similar after the conversation than before and, when negative, indicates that they were more similar before the conversation than after the conversation. We mean-centered and z-scored participants' trait survey responses before calculating alignment to account for participant-level biases in scale use.

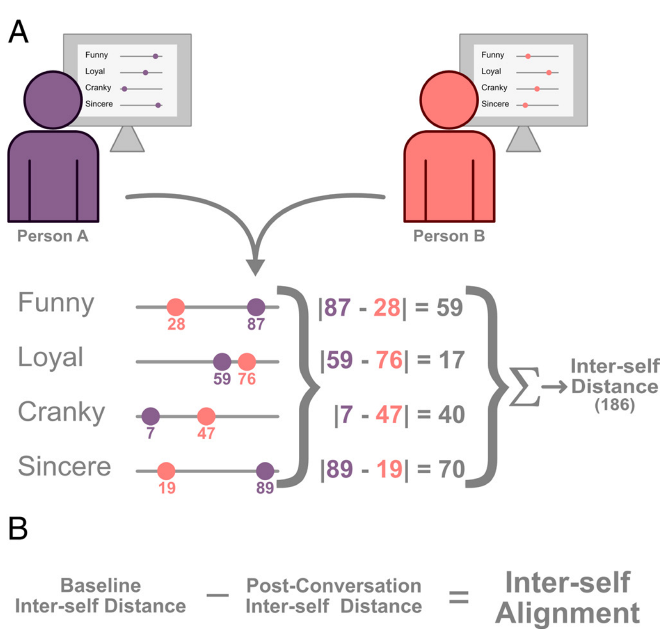

Fig. 2. Schematic of how inter-self distance and alignment were calculated. (*A*) Participants filled out 60-item trait scales from (56). In this illustrated example, we use four traits taken from the scale. To compute the distance between pairs' trait ratings, we calculated the sum of the absolute value of the differences between the two survey responses (i.e., the Manhattan distance). Responses were mean-centered and z-scored for each participant prior to distance calculation to account for any idiosyncratic differences in scale usage. (*B*) Inter-self alignment was calculated by subtracting the distance between partners' trait ratings after the conversation from the distance between their trait ratings at baseline. Positive values indicate that participants' self-views became more similar after the conversation and negative values indicate that participants' self-views became less similar after the conversation.

Conversation enjoyment. Two questions from the post-conversation survey were related to conversation enjoyment: "How much did you enjoy the conversation that you just took part in?" and "If you were to see the person you just interacted with somewhere on campus, how likely would you be to stop and have a conversation with them?" Participants responded to both questions using a sliding scale from 0 (Not at all) to 100 (Extremely) with a midpoint at 50 (Moderately). We first examined the relationship between these two variables in a linear mixed effects model with random intercepts for the participant, conversation partner, dyad, and group. We found a significant, positive relationship between the two variables [$\beta = 0.66$, 95% CI = (0.57,0.75), $P < 0.001$] and so took their average to be an index of conversation enjoyment. These steps were detailed in our preregistration.

Self-reported closeness. Three questions from the post-conversation survey measured participants' perceived closeness with their conversation partner: "How emotionally close do you feel to the person you just talked with?", "How similar, in terms of personality, temperament, major likes and dislikes, beliefs, and values, do you feel to the person you just talked with?", and the inclusion of the other in the self scale (62). Participants responded to the first two questions with a sliding scale from 0 (Not at all) to 100 (Extremely) with a midpoint at 50 (Moderately) and followed standard protocol for the inclusion of the other in the self scale.

After confirming that the data were adequate for dimensionality reduction [Bartlett's test of sphericity $\chi^2(3) = 263.78$, $P < 0.001$; Kaiser–Meyer–Olkin test KMO = 0.695], we mean-centered and z-scored the data and ran a PCA on the three questions related to perceived closeness. The elbow of the scree plot determined that one component was an appropriate approximation of the three variables, capturing a large amount of the shared variance between them (70.8%). We therefore used the first principal component of the PCA as a composite self-reported closeness score. Because inter-self alignment was not significantly related to perceived closeness (*SI Appendix*, Supplementary Materials), we do not focus on this variable in the results below. Interested readers can refer to *SI Appendix*, Supplementary Materials to learn more about which variables were related to perceived closeness.

Partner prediction accuracy. To assess the accuracy of participants' predictions of their conversation partners' trait ratings, we found the Manhattan distance between participants' ratings of their partners' traits and their partners' ratings of their own traits after their conversation. We then reversed the sign of the distances so that larger values would indicate greater partner prediction accuracy. Both sets of ratings were mean-centered and z-scored before distances were calculated.

Normative self-views. To estimate the normative self-view in our sample, we found the average rating of each trait across all participants at baseline. Following the procedures we used for calculating inter-self alignment, trait ratings were z-scored and mean-centered prior to the averaging.

Results

All analyses were carried out in R (version 4.2.2). Linear mixed effects models were implemented using the *lme4* package and were fit with maximum likelihood estimation; degrees of freedom and p-values were approximated using Satterthwaite's method via the *lmerTest* package (63, 64). We report standardized regression coefficients to increase interpretability.

Manipulation Checks and Scale Validation.

Conversation prompt effectiveness. Overall, most participants reported discussing all of their three assigned conversation prompts (mean = 2.81, median = 3, SD = 0.61). We used a linear mixed effects model to assess whether the assigned prompts (deep or shallow) influenced the reported depth of participants' conversations. Random intercepts were included for each participant, conversation partner, dyad, and group to account for repeated measurements resulting from the round-robin design. The analysis revealed a significant, positive relationship between assigned topic and reported conversation depth [$\beta = 0.47$, 95% CI = (0.31,0.62), $P < 0.001$], indicating that participants in the deep topics condition reported having deeper conversations than those in the shallow topics condition.

Trait scale validation. The traits in the scale used here have been used in multiple studies investigating self- and other-knowledge (56, 65, 66). We ran analyses to confirm that the scale reliably captured our participants' unique views of themselves. For example, if a participant considers themself to be funny, then they should consistently rate themself as funny every time they fill out the same scale. The extent to which they see themself as funny may fluctuate with each conversation, but we expect that they will have relatively stable self-views. To verify that subjects' responses to our scale were indeed consistent within each subject, we compared the distances between subjects' five scale responses (baseline, after each conversation, follow-up) to the distances between responses from different subjects, expecting distances to be smaller within subjects than between subjects.

We created a 460 × 460 model matrix with 5 × 5 squares along the diagonal set equal to 1 and the remainder of the matrix set to 2 (*SI Appendix*, Fig. S2, *Left*). Each of the 5 × 5 squares (n = 92) represented the distances between the five measurements of a single subject (at baseline, after each conversation, and at follow-up). Four participants who did not complete the follow-up survey were excluded from this analysis. We compared this model matrix to an equivalent matrix of Manhattan distances between subjects' trait scale responses (*SI Appendix*, Fig. S2, *Center*) with a Mantel test. The test revealed a significant fit between our model and the observed data ($r = 0.09$, $P < 0.001$), suggesting that within-subject distances were significantly smaller than between-subject distances (*SI Appendix*, Fig. S2, *Right*). This result implies that the trait scale reliably measures self-views.

General Self-View Alignment. We first assessed, across all subjects, whether self-views converged during conversation. To determine whether conversations correspond with participants' self-views becoming more similar to their partners' self-views, we compared the distribution of observed alignment values to 0 using a one-sample *t* test. The distribution of alignment values fell significantly above 0 [$M_{alignment} = 6.77$, 95% CI = (4.94, 8.59), $t(143) = 7.34$, $P < 0.001$; Fig. 3*A*], suggesting that most pairs' self-views became more similar after the conversation compared to before the conversation. The distribution of inter-self alignment scores remained significantly greater than zero [Intercept = 6.85, 95% CI = (4.58, 9.11), $P < 0.001$] in a linear mixed effects model which, in addition to the intercept term, included predictors for baseline inter-self distance, assigned depth of conversation topics, and prior familiarity with their conversation partner as well as random intercepts for each group. The dependent variable, inter-self alignment, was not mean-centered or z-scored in this model in order to estimate its mean with the intercept term. Assigned topic depth was not a significant predictor of inter-self alignment in this model [$\beta = -0.81$, 95% CI = (−3.10, 1.49), $P = 0.488$], suggesting that conversation partners did not align more in the deep condition than in the shallow condition. Additionally, greater baseline inter-self distance predicted greater inter-self alignment [$\beta = 3.99$, 95% CI = (2.10, 5.88), $P < 0.001$]. This relationship was expected given that, if two people are farther apart in trait space to begin with, they are able to move a greater distance toward each other than two people who started next to each other. Given this relationship, we include baseline inter-self distance as a control variable in all models which also include our measure of inter-self alignment. The distribution of alignment values fell significantly above zero, indicating that subjects made more similar trait ratings after their conversations than before.

As a test of the robustness of this effect, we ran a permutation-based analysis in which we compared the observed median value of inter-self alignment ($Med_{alignment} = 7.46$) to a distribution of 5,000 median alignment values obtained by randomly shuffling the order of traits in the inter-self alignment calculation. The observed median fell well above the distribution of medians from permuted values (maximum permuted median = 3.75;

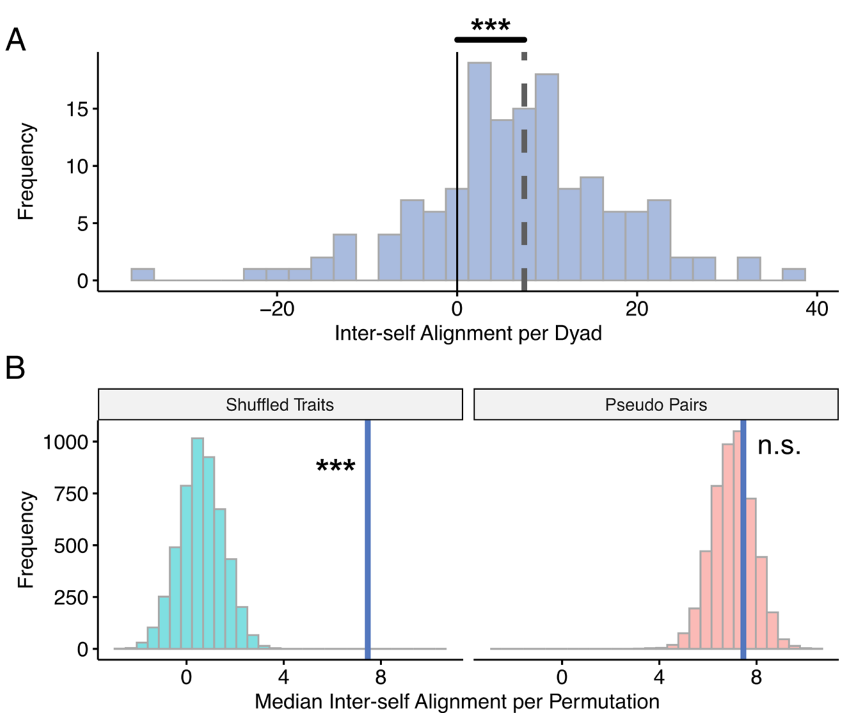

Fig. 3. (*A*) Histogram depicting the distribution of inter-self alignment scores per dyad. The dashed gray line represents the mean of the distribution, the solid black line marks 0, and *** denotes $P < 0.001$. (*B*) Histograms showing null distributions of median inter-self alignment values generated from shuffling the order of traits before calculating alignment (blue/*Left*) or from distributions of alignment values generated from pseudopairs (red/*Right*). Each null distribution is made of 5,000 permuted values. The observed median (solid purple line) falls above the former, but not the latter null distribution.

Fig. 3*B*), once again suggesting that conversations aligned all pairs' self-views.

Next, we asked whether conversation-induced alignment was specific to dyadic sets. If a unique, cocreated self-view emerges during each conversation, then inter-self alignment effects should be stronger within-versus between-dyadic pairs. To assess this, we compared the median inter-self alignment value observed in our real pairs of dyads to a distribution of 5,000 median inter-self alignment values generated from randomly assigned pseudopairs. Inter-self alignment was calculated for the pseudopairs exactly as it was for real pairs, except for each iteration of pseudopairs, dyads consisted of randomly assigned participants. The observed median value of inter-self alignment fell in the middle of the permuted distribution with 30% of the permuted distribution being greater than the observed value (maximum permuted median = 10.55; Fig. 3*B*). This indicates that all participants in the study became more aligned after their conversations as compared to before.

Overall, significant results from the comparison to 0 and shuffled traits suggest that conversation tends to align self-views. However, results from the comparison to pseudoconversation pairs suggest that alignment in a conversation occurs in a general way across conversations, rather than in a dyad-specific way.

Notably, we also found that self-views tended to remain aligned at follow-up as compared to baseline. We built a linear mixed effects model with dyad-level inter-self distance values as the dependent variable and a categorical time variable (baseline, post-conversation, and follow-up) as our predictor. The model also included random intercepts for each dyad and group. We excluded data from any dyad with a participant who did not fill out the follow-up survey (N=12). There was a significant effect of time on inter-self distance ($F(2,264) = 25.84$, $P < 0.001$; *SI Appendix*, Fig. S5). Post hoc Holm-corrected pairwise contrasts revealed that inter-self distances at post-conversation ($M = 36.53$, $t(266) = 6.42$, $P < 0.001$) and follow-up ($M = 37.04$, $t(266) = 5.96$, $P < 0.001$) were significantly smaller than baseline ($M = 43.58$), indicating relatively persistent increases in similarity.

Inter-self Alignment Predicts Greater Conversation Enjoyment. To test whether greater inter-self alignment predicts greater conversation enjoyment, we built a linear mixed effects model with one observation for each participant in each conversation ($n = 288$). The index of conversation enjoyment was included as the outcome variable, and inter-self alignment was our predictor of interest. We included the conversation topic (deep or shallow), the interaction between topic and inter-self alignment, and self-reports of how well the participants knew each other before the study as covariates. Since the similarity of a dyad's self-views at baseline limits the extent to which they can align, we also controlled for participants' baseline inter-self distance. All continuous predictors were mean-centered and z-scored. To account for the repeated measurements inherent in a round-robin design, we included random intercepts for each participant, partner, dyad, conversation number (an ID variable coding for each participant's first, second, or third conversation), and group.

As predicted, we found a significant, positive effect of inter-self alignment on conversation enjoyment [$\beta = 0.18$, 95% CI = (0.05, 0.30), $P = 0.007$; Fig. 4*A*]. In addition, the distance between partners' baseline self-views [$\beta = -0.17$, 95% CI = (−0.31, −0.03), $P = 0.020$] and how well the participants knew each other before the study [$\beta = 0.28$, 95% CI = (0.18, 0.38), $P < 0.001$] were also significantly related to conversation enjoyment. Contrary to our predictions, topic depth did not significantly interact with inter-self alignment to predict enjoyment [$\beta = -0.01$, 95% CI = (−0.13, 0.11), $P = 0.874$], indicating that even shallow conversations can produce inter-self alignment effects.

The relationship between inter-self alignment and conversation enjoyment is ephemeral. To examine the duration of the association between inter-self alignment and enjoyment, we ran the same model again, except that inter-self alignment was calculated using inter-self distances from the baseline (pre-conversation) survey and the follow-up survey that took place one week after their final conversation. The linear mixed effects model had identical random effects and covariates as the original model and the only significant predictor was the self-report of how well the participants knew each other before the study [$\beta = 0.25$, 95% CI = (0.12, 0.37), $P < 0.001$]. The extent to which conversation partners' trait ratings became more similar over the course of the entire study (i.e., from their pre-conversation baseline to a week after their final conversation) was not related to how much they enjoyed their conversation [$\beta = -0.01$, 95% CI = (−0.15, 0.14), $P = 0.907$]. Instead, each dyad's enjoyment was solely predicted by the degree to which their self-views became more aligned immediately after their conversation.

The relationship between inter-self alignment and conversation enjoyment manifests on a dyadic level. The results above suggest that the more conversation partners align their self-views during a conversation, the more they enjoy talking with one another. Next, we asked whether this effect is driven by something specific individuals bring to all their conversations versus something attributable to specific conversation partners (i.e., dyads). In other words, do some people tend to align with all their conversation partners and does that trait-like tendency predict enjoyment? Or, is there something special about two particular people conversing

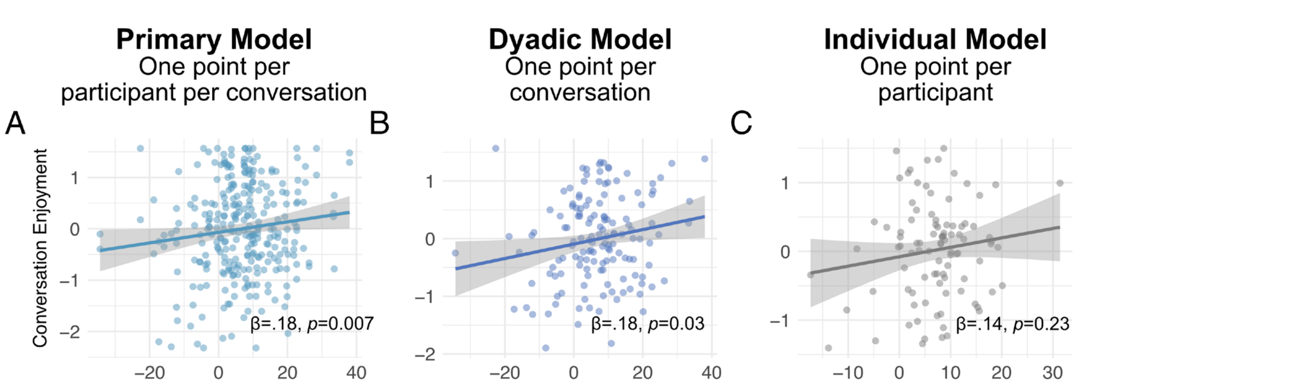

Fig. 4. Scatter plots with regression lines showing the relationship between conversation enjoyment and inter-self alignment with each member of each conversation (*A*), each dyad (*B*), and each participant (*C*) treated as a single observation.

in ways that align their self-views that predicts more enjoyment for both of them?

Leveraging the round-robin structure of our data, we were able to determine whether inter-self alignment was related to conversation enjoyment at the level of the dyad and/or the individual. To do so, we averaged all relevant variables (i.e., conversation enjoyment, inter-self alignment, baseline inter-self distance, self-reports of how well participants knew each other prior to the study, assigned topic, and identification variables) within conversation pairs for the dyadic analysis and within each participant for the individual analysis. We then built linear mixed effects models with identical fixed effects structures as the original model. The random effects differed from the original model; both the dyadic and individual models only included random intercepts for each round-robin group since the unit of analysis became coarser in each model.

In the dyadic model, inter-self alignment was a significant positive predictor of conversation enjoyment [$\beta = 0.18$, 95% CI = (0.02, 0.34), $P = 0.03$; Fig. 4*B*]. In contrast, inter-self alignment was not a significant predictor of conversation enjoyment in the individual model [$\beta = 0.14$, 95% CI = (−0.08, 0.36), $P = 0.23$; Fig. 4*C*]. Baseline inter-self distance [$\beta_{dyad} = -0.23$, 95% CI = (−0.41, −0.05), $P = 0.01$; $\beta_{individual} = -0.25$, 95% CI = (−0.52, −0.01), $P = 0.04$] and how well the participants reported knowing each other before the study [$\beta_{dyad} = 0.39$, 95% CI = (0.24, 0.53), $P < 0.001$; $\beta_{individual} = 0.32$, 95% CI = (0.12, 0.53), $P = 0.001$] were the only other significant predictors in both models. Thus, the relationship between inter-self alignment and conversation enjoyment appears to be driven by properties of conversation pairs, not single individuals. Interested readers can refer to *SI Appendix*, Supplementary Materials to see complementary analyses investigating individual-level effects of conformity and influence.

Inter-self Alignment, Enjoyment, and Partner Prediction Accuracy. Results reported above demonstrate that when self-views converge during conversation, the conversation is more enjoyable than when self-views do not converge. However, these results alone do not reveal the underlying cognitive mechanism that may link aligned self-views and enjoyment. Based on prior research, we wondered whether an underlying mechanism may be prediction: specifically, whether aligning self-views affords more accurate predictions of one's partner which, in turn, make conversations more enjoyable. This would be consistent with three bodies of past work indicating that 1) when we mentally simulate others and make predictions about them, we incorporate information about them into our own self-views (52, 56, 66), 2) accurate predictions are rewarding (67), and 3) making more accurate predictions about social partners has been linked to both behavioral adaptability (adapting one's behavior to fit a partner's preferences) and having higher-quality social relationships (68, 69).

To test this possibility, we investigated whether conversation partners who demonstrated greater inter-self alignment were also more accurate in predicting each other's traits. We quantified a participant's accuracy by calculating the distance between 1) participants' ratings of their partner's self-views and 2) their partner's ratings of their own self-views after the conversation, here termed "other-accuracy" (Fig. 5*A*). We reversed the sign of these distances so that larger values indicated greater accuracy when predicting a partner's trait ratings. Note that "other-accuracy" is not mathematically redundant with inter-self alignment. Someone could accurately perceive that their partner is sincere—leading to high other-accuracy—but not become any more sincere themselves over the course of their conversation—limiting their inter-self alignment.

We constructed a linear mixed effects model with other-accuracy as the outcome variable, inter-self alignment as the predictor of interest, and baseline inter-self distance, assigned conversation topic, and the familiarity with the conversation partner as covariates. We included random intercepts for each participant, conversation partner, dyad, conversation number, and group to account for repeated measurements. All variables were mean-centered and z-scored. Greater inter-self alignment significantly predicted other-accuracy [$\beta = 0.65$, 95% CI = (0.56, 0.74), $P < 0.001$; Fig. 5*B*]. Greater inter-self distance at baseline also predicted less other-accuracy [$\beta = -0.51$, 95% CI = (−0.61, −0.41), $P < 0.001$].

Other-accuracy also predicted greater conversation enjoyment [$\beta = 0.32$, 95% CI = (0.20, 0.43), $P < 0.001$; Fig. 5*B*] in a linear mixed effects model controlling for inter-self distance at baseline, assigned conversation topic, and prior familiarity with the conversation partner. All terms were mean-centered and z-scored. Random intercepts were included for each participant, partner, dyad, conversation number, and group. Prior familiarity with the conversation partner also predicted greater conversation enjoyment [$\beta = 0.27$, 95% CI = (0.17, 0.36), $P < 0.001$].

Given that other-accuracy related to both inter-self alignment and conversation enjoyment, we ran a mediation analysis with inter-self alignment as our independent variable, other-accuracy as a mediator, and conversation enjoyment as our dependent variable, using the R package *mediation* (70). All regressions included controls for baseline inter-self distance, conversation topic, and prior familiarity with the conversation partner as well as a random intercept for each participant.[*] All variables were mean-centered and z-scored and we used nonparametric bootstrap estimation with 5,000 Monte Carlo draws to estimate CI and *P* values. Other-accuracy fully mediated the relationship between inter-self alignment and conversation enjoyment [ACME = 0.26, 95% CI = (0.15, 0.38), $P < 0.001$; ADE = −0.06, 95% CI = (−0.21, 0.11), $P = 0.53$; Fig. 5*C*].

Exploring How Self-views Align Using the 3D Mind Model. Our results demonstrate three things: 1) peoples' views of themselves tend to align in conversations, 2) alignment happens more during enjoyable conversations, and 3) the relationship between alignment and enjoyment is mediated by how well participants can predict their partners' self-views. We have not yet examined whether specific aspects of the self are more affected by conversation. Just as some mental states we experience, such as anger or sleepiness, are more or less likely to change from moment to moment, some aspects of the self might be more malleable in conversations (71–74). Similarly, aligning self-views in certain ways may be more predictive of conversational enjoyment. To examine these questions, we used the 3D Mind Model—a well-established model of mental states—to reduce our trait space into three dimensions: valence, social impact, and rationality (75–77). The dimension of valence describes the extent to which mental states are either positive or negative, social impact describes the extent to which states are both intense and socially directed, and rationality describes the extent to which states are cognitive or emotional (78).

The *affectr* R package allowed us to get weights for each dimension of the 3D Mind Model for all of our 60 traits (79); *SI Appendix*, Fig. S8 for a heatmap of entire weights matrix). Fig. 6 shows four example traits and their weights for each dimension of the 3D Mind Model. We then used these weights to project participants' responses to the 60 traits into a three-dimensional

[*]The R package only allows for the inclusion of one random effect, so we elected to include the one that routinely explains the most variance in the other reported mixed effects models (participants).

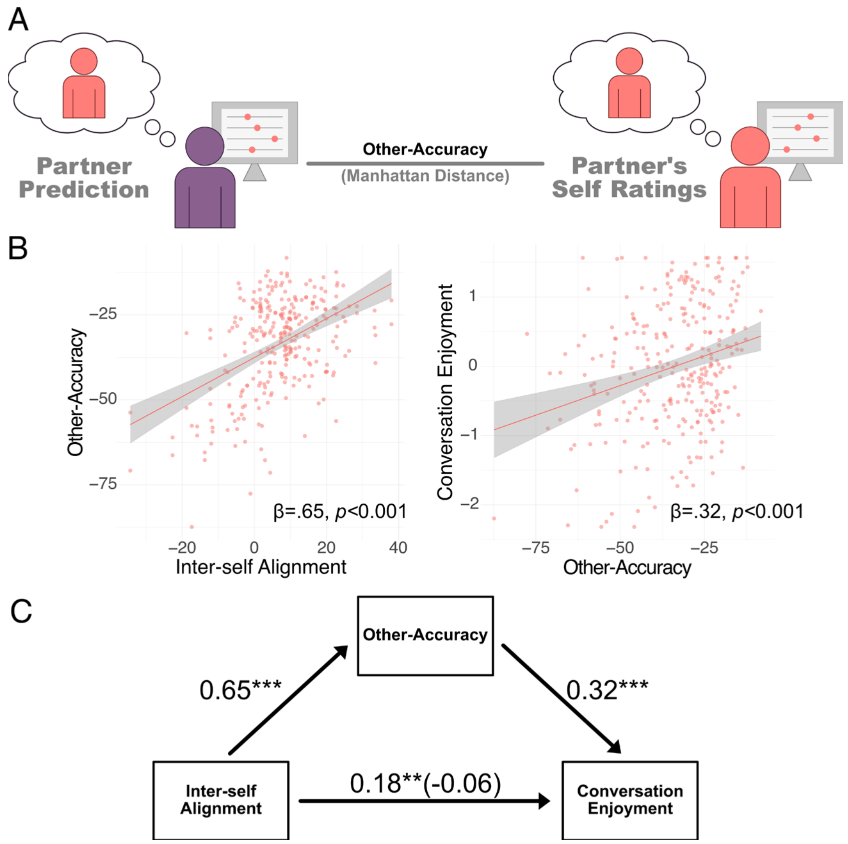

Fig. 5. (*A*) Schematic of how partner prediction accuracy (or "other-accuracy") was calculated. We found the Manhattan distance between a participant's rating of their partner's traits to their partner's rating of their own traits. Distances were reversed scored so that larger values indicated greater accuracy in predicting their partner's trait ratings. Trait ratings were mean-centered and z-scored for each participant prior to distance calculation to account for variations in scale usage across participants. (*B*) Scatter plots with regression lines showing the relationship between (*Left*) inter-self alignment and other-accuracy and (*Right*) other-accuracy and conversation enjoyment. (*C*) Depiction of how other-accuracy fully mediates the relationship between inter-self alignment and conversation enjoyment. Standardized regression coefficients are shown next to their corresponding arrows. The coefficient in parentheses reflects the relationship between inter-self alignment and conversation enjoyment when controlling for other-accuracy. All models control for baseline inter-self distance, reported prior familiarity with the conversation partner, and depth of assigned topics. Random intercepts were included for each participant. * signifies $P < 0.05$, ** signifies $P < 0.01$, and *** signifies $P < 0.001$.

space by multiplying a matrix of participants' trait scale responses by the 3D Mind Model weights matrix. This multiplication resulted in a new matrix with three 3D Mind Model scores for each participant which were used for a new set of exploratory analyses. As before, participants' trait ratings were mean-centered and z-scored prior to the multiplication to account for differences in scale usage across participants.

Self-views align on social impact and rationality. To calculate alignment on each dimension of the 3D Mind Model, we found the absolute value of the difference between conversation partners' scores on each dimension in the baseline survey and in the post-conversation survey. These absolute differences describe the distance between dyads' scores on each dimension. We then subtracted the distance between dyads' scores after the conversation from the distance between their scores before the conversation to gauge how much each dyad aligned on each dimension of the 3D Mind Model. The resulting values are larger if conversation partners had more similar scores on a given dimension after the conversation compared to before the conversation.

Initially, to gauge whether pairs were more likely to align on certain dimensions of the 3D Mind Model, we compared the distribution of alignment values for each dimension to 0 with one-sample t tests. Only rationality and social impact showed significant alignment across most participants [$M_{rationality} = 0.59$, $t_{rationality}(143) = 3.76$, 95% $CI_{rationality} = (0.28\ 0.91)$, $p_{rationality} < 0.001$; $M_{social\ impact} = 0.37$, $t_{social\ impact}(143) = 2.56$, 95% $CI_{social\ impact} = (0.08,\ 0.66)$, $p_{social\ impact} = 0.011$] and these tests remained significant after multiple comparisons correction via the Holm–Bonferroni method (80).

To gauge whether these effects were general across all participants or dyad specific, we compared the observed medians of alignment on each dimension to null distributions of 5,000 medians of alignment calculated from combinations of pseudopairs. Alignment on all dimensions fell within the null

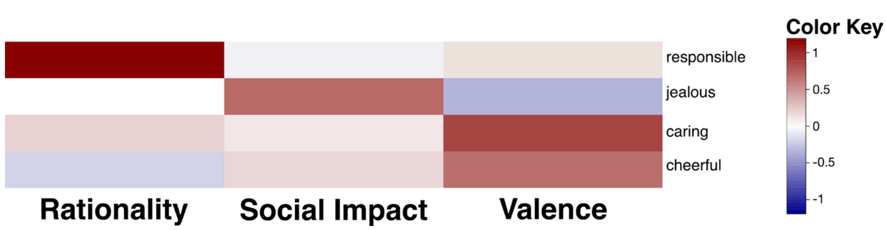

Fig. 6. Heatmap showing 3D Mind Model weights for four example traits. *SI Appendix*, Fig. S8 for a heatmap of the entire weights matrix.

distributions (13% of null results above median of rationality, 74% of null results above median of social impact, and 20% of null results above median of valence). These results suggest that all pairs tended to align on dimensions of the self-related to rationality and social impact, but that this was a general trend observed across all participants in the study.

Alignment on social impact and valence is predictive of enjoyment. Conversation partners who become more aligned in their self-views during a conversation also have more enjoyable conversations, but do all aspects of the self matter? If one dyad's conversation leads them to see themselves as funnier and another dyad's conversation leads them to feel more independent, will both dyads experience an equivalent boost in enjoyment? Or, are certain aspects of inter-self alignment more predictive of enjoyment than others? To answer these questions, we investigated the relationship between alignment on each dimension of 3D Mind Model space and conversation enjoyment.

To validate this dimensionally reduced space and check the robustness of our main findings, we reran our model relating inter-self alignment to conversation enjoyment using participants' 3D Mind Model scores in place of their trait ratings. We built linear mixed effects models with identical fixed and random effects structures as the original model, except that inter-self alignment and baseline inter-self distance were calculated in 3D Mind Model space. Replicating our earlier results, participants whose self-views became more similar to their partners' tended to have more enjoyable conversations [$\beta = 0.22$, 95% CI = (0.08, 0.36), $P = 0.002$]. Baseline distance between conversation partners' positions in 3D Mind Model space [$\beta = -0.15$, 95% CI = (−0.29, −0.01), $P = 0.03$] and self-reports of how well the conversation partners knew each other before the study [$\beta = 0.28$, 95% CI = (0.18, 0.38), $P < 0.001$] were also significant predictors of enjoyment.

To test whether alignment on any of the three 3D Mind Model dimensions was predictive of conversation enjoyment, we related alignment on each dimension to conversation enjoyment in three separate linear mixed effects models, controlling for baseline distance in 3D Mind Model space, assigned conversation topic, and self-reports of how well they knew their partner before the study. The models also included random intercepts for each participant, conversation partner, dyad, group, and conversation number. Alignment on social impact [$\beta = 0.15$, 95% CI = (0.02, 0.28), $P = 0.02$] and valence [$\beta = 0.17$, 95% CI = (0.04, 0.31), $P = 0.01$] were significant predictors of conversation enjoyment, and both remained significant after multiple comparison corrections using the Holm–Bonferroni method (80). Based on these results, we reran our mediation analysis two more times with alignment and other-accuracy generated from valence and social impact scores. Both models returned comparable results to the original model [$ACME_{social\ impact} = 0.08$, 95% CI = (0.03, 0.15), $P < 0.001$; $ADE_{social\ impact} = 0.06$, 95% CI = (−0.07, 0.18), $P = 0.374$; $ACME_{valence} = 0.12$, 95% CI = (0.03, 0.21), $P = 0.004$; $ADE_{valence} = 0.13$, 95% CI = (−0.02, 0.29), $P = 0.111$].

Inter-self Alignment Corresponds with Developing Normative Self-views. Finally, we assessed whether inter-self alignment in a given conversation may occur, at least in part, to more generally homogenize participants' self-views relative to community norms. This possibility fits with prior social influence research showing that dyadic conversations can facilitate the alignment of beliefs not just between dyads, but also across social networks (35). First, we calculated the normative self-view in our sample by averaging all participants' scaled trait ratings at baseline. Baseline self-views served as the best estimation of the norm since they were collected before any study-related conversations could have influenced them. We then computed the Manhattan distance between participants' self-views at each time point in the study and the normative self-view (Fig. 7*A*). As in prior analyses, trait ratings were mean-centered and z-scored prior to distance calculation. To determine whether participants tended to become more normalized after each conversation, we built a linear mixed effects model with time point labels (i.e., "Baseline," "Conversation 1," etc.) as our independent variable, distance to the normative self-view as the outcome variable, and random intercepts for each participant. We applied four orthogonal polynomial contrasts to the time point labels and found that both the linear ($\beta = -0.23$, 95% CI = (−0.36, −0.10), $P < 0.001$] and quadratic [$\beta = 0.24$, 95% CI = (0.11,0.37), $P < 0.001$] contrasts were significant, suggesting that self-views became more similar to the sample norm over time, but that this trend weakened once a full week had passed since the conversations (Fig. 7*B*). To ensure that the increase in normativeness was driven by the conversations themselves (rather than a difference at the one-week follow-up), we further assessed whether data from the three conversations alone drove the decreasing linear trend in distance from the norm. To this end, we built a model equivalent to the earlier model, except that it a) only included data from the three post-conversation surveys and b) only included linear and quadratic contrasts. In this model, just the linear contrast was significant [$\beta = -0.12$, 95% CI = (−0.22, −0.03), $P = 0.013$], indicating increased normativeness with more conversation. Overall, these results suggest that the three conversations normalized participants' self-views.

Next, we asked whether this shift toward normativeness was related to inter-self alignment. In addressing this question, we first calculated change scores for normativeness to mirror the calculations underlying inter-self alignment. For each participant in each conversation, we subtracted their post-conversation distance from the normative self-view from their pre-conversation distance from the normative self-view, resulting in a value that is positive if a participant moved closer to the norm after their conversation and is negative if they moved farther away. These change scores were strongly correlated with participants' inter-self alignment scores ($r = 0.60$, $P < 0.001$), suggesting that inter-self alignment corresponds with a shift toward more normative self-views.

These results suggest that inter-self alignment and changing toward the normative self-view are closely related processes. Might this mean that inter-self alignment in one-off conversations helps serve a broader function to shift self-views into alignment with community norms? If so, then inter-self alignment should specifically act on participants' *non-normative* self-views to help bring them closer to the community's norm. To test this possibility, we calculated trait-wise inter-self alignment scores for each participant in each conversation. These scores tell us the degree to which conversation partners' ratings of each trait became more similar after the conversation as compared to before the conversation. In addition, we calculated trait-wise normativeness scores for each participant at baseline. These scores tell us how far each participant's ratings of each trait fell from the norm at baseline. For each participant in each conversation, we then correlated their trait-wise inter-self alignment scores and their baseline trait-wise normativeness scores. Positive correlations indicate that participants moved closer to their partner on the traits on which they fell farther from the norm at baseline. Negative correlations indicate that participants aligned more with their partner on their more normative self-views. While the distribution of correlation values included both positive and negative values, a Wilcoxon signed-rank test demonstrated that it fell significantly above zero with a median r of 0.26 ($Z = 14.03$, $P < 0.001$; Fig. 7*C*). Thus, participants' least normative self-views tended to be those that shifted most to align with their conversation partner. Put another way, if a person did

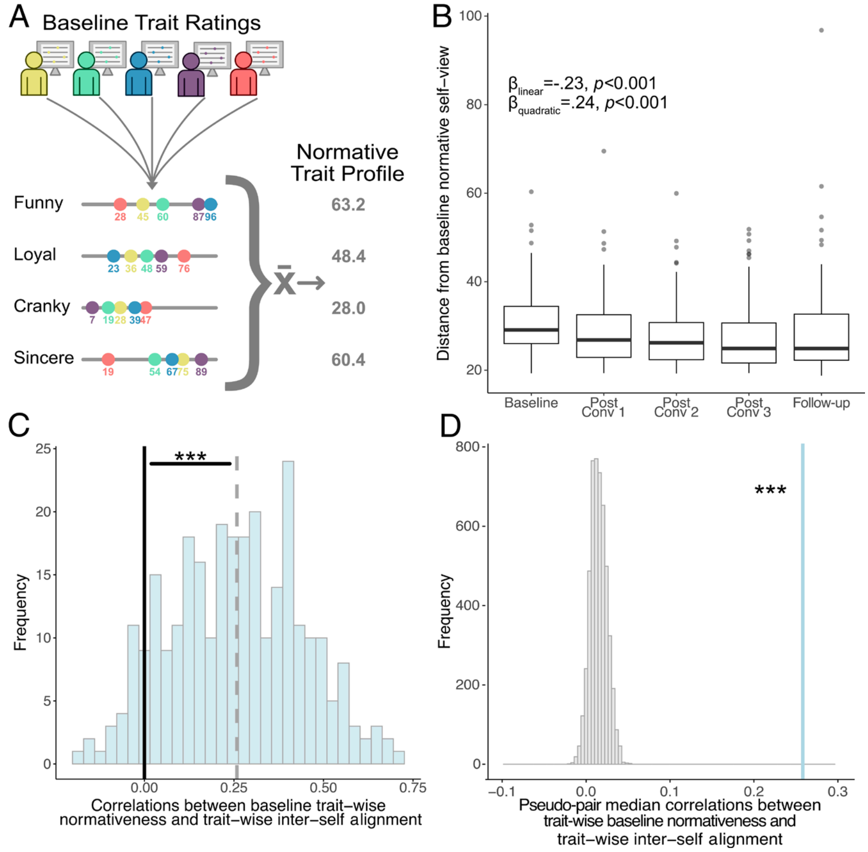

Fig. 7. (*A*) Depiction of how the normative trait profile was calculated. All participants' baseline ratings for each trait were averaged to create a vector representing the sample's normative trait profile. Trait ratings were mean-centered and z-scored prior to averaging. (*B*) Boxplots showing the distance from participants' trait ratings to the community norm over time. Both linear and quadratic contrasts over time were significant, suggesting that participants tended to become more normalized throughout the study, though this trend attenuated one week later. (*C*) Histogram showing correlation values between trait-wise normativeness values and inter-self alignment values. For each participant's rating of each trait at baseline, we calculated how far that rating fell from the normative trait profile. These values represent each participant's trait-wise normativeness scores. Then, for each conversation, we determined the extent to which their ratings for each trait became more similar to their partner's rating after the conversation as compared to before the conversation. These values represent the trait-wise inter-self alignment scores. Finally, for each participant, we correlated these two vectors. The plot in (*C*) shows the distribution of those values and demonstrates that the distribution falls significantly above zero ($Z = 14.03$, $P < 0.001$). The dashed line shows the median of the distribution and the solid line shows 0. (*D*) We then compared the median from (*C*) to a null distribution of median correlation values obtained from pseudopair comparisons. In other words, we shuffled participant identities when performing the correlation analysis to determine whether the relationship between trait-wise normativeness and trait-wise inter-self alignment was idiosyncratic. Since the real median (solid blue vertical line) falls far above the null distribution, we can conclude that the relationship is participant specific.

not view themselves on the trait "funny" similarly to how the whole sample did at baseline, they were more likely to align with their partner on that trait. Inter-self alignment might serve the broader function of bringing a community's self-views into alignment.

Finally, we tested whether these relationships were person specific. It is possible that all dyads align on a common set of traits which incidentally brings most people's non-normative self-views toward the norm. For example, having a conversation might simply make people feel more social or outgoing which would normalize self-views by bringing all conversation partners toward the norm on those traits. In contrast, inter-self alignment might target the unique ways in which participants see themselves as non-normative. To arbitrate between these possibilities, we compared the median correlation value from the prior test to 5,000 median correlation values generated from pseudopairs in which identities for trait-wise alignment values were randomly shuffled, so that one participant's trait-wise alignment scores were randomly matched with another participant's trait-wise normativeness scores. We found that the real correlations between trait-wise alignment and normativeness scores fell far above the permuted values (maximum median from permuted distributions = 0.05; Fig. 7*D*). The striking difference between the real and permuted medians suggests that the relationship between alignment and normativeness is idiosyncratic. Inter-self alignment appears to bring people closer to average on their unique set of non-normative traits.

Discussion

It is well established that talking with other people can change the way we think about the world (15–17, 19–22, 81. Here, we demonstrate that conversation even changes the way we think about ourselves. Specifically, we show that conversation partners tend to think of themselves more similarly after a conversation than before, an effect we term "inter-self alignment." Further, the more partners' self-views aligned, the more they enjoyed their conversation. This effect manifested on a dyadic level, but not an individual level. An individual who tended to align with each of their conversation partners did not necessarily enjoy each of their conversations. Instead, enjoyable conversations were those in which *both* conversation partners became aligned with each other. Finally, inter-self alignment moves individuals toward normative self-views by making each partner's self-views less idiosyncratic.

Contrary to our predictions, the relationship between inter-self alignment and enjoyment was not specific to deep conversations but manifested equally across both shallow and deep conversation conditions. This suggests that inter-self alignment does not require conscious reflection on the self but may be occurring in the background, consistent with spontaneous activity in self-relevant neural regions during mind-wandering and at rest (82–85).

The relationship between inter-self alignment and enjoyment was further mediated by how well participants were able to estimate their partners' own self-views. This suggests that developing more similar self-views over the course of a conversation is associated with having a better understanding of one's partner, leading to more enjoyable conversations. Exploratory analyses demonstrated that some aspects of the self converged more than others. Using the 3D Mind Model (76), we found that self-views tended to align on the dimensions of social impact and rationality, while aligning on the dimensions of social impact and valence predicted enjoyment. Social impact—capturing the degree to which trait ratings are socially directed (e.g., high ratings of adjectives like "jealous," "friendly," or "passionate")—was the only dimension that significantly demonstrated alignment and predicted conversation enjoyment. This suggests that conversations are most likely to facilitate social connection when people become a similarly social self, echoing the classic William James quote that "a man has as many social selves as there are individuals who recognize him"(86).

Inter-self alignment persisted for at least a week (the last survey given), suggesting that even short conversations with strangers may have lingering effects on our self-views. This suggests that more frequent interactions could have a more enduring impact on self-views, consistent with research on friends becoming more similar over time (30). If so, the convergence of self-views through repeated conversation may be one mechanism underlying perceptions of self-other overlap (61, 62, 87, 88). This persistence of inter-self alignment was true regardless of how enjoyable the conversations were at the time. Although enjoyment predicted inter-self alignment at the time of the conversation, it did not predict alignment a week later. This may be due, in part, to the fact that each pair of conversation partners only interacted once. In contrast, repeated interactions have been found to induce feelings of connectedness and well-being (51, 89). Even if each individual conversation in our study evokes an ephemeral change in affect, having a high quantity of enjoyable interactions could profoundly affect feelings of connectedness and well-being in the aggregate (90, 91).

We also found evidence that inter-self alignment facilitates the normalization of self-views; conversation appears to make people particularly likely to view themselves more normatively by disproportionately adjusting their idiosyncratic self-views. This could contribute to the development of homophily in social networks (25, 92, 93. While the literature on homophily primarily views the development of homophily as a selection process (23, 24, 94), our work—along with recent findings from van Zalk et al. (30)—suggests that we may also become more homophilous through conversation. Further investigation is needed to determine the extent to which the effects reported here compound at longer timescales and between existing relationship partners (e.g., friends; romantic relationships).

The normativeness results have interesting implications for recent work demonstrating that lonely individuals and peripheral members of social networks have more idiosyncratic responses to social stimuli (95–98). For example, although people tend to show similar neural responses to videos from popular media (22, 26, 99), lonelier individuals, as well as individuals in objectively peripheral social network positions, show more atypical neural responding (95, 96). Likewise, loneliness corresponds with idiosyncratic communication about and neural representations of shared cultural knowledge (97). Particularly relevant to our findings, lonely individuals' neural representations of themselves are dissimilar from their neural representations of other social network members (98). This suggests lonely individuals may come into conversations with self-views that fall farther from the norm. Additionally, because of their isolation, lonely individuals may have fewer conversational opportunities to move themselves closer to the norm. Future work is needed to assess whether and how conversations may explain and/or attenuate feelings of isolation, in part, by bringing self-views and perceptions of the social environment more generally, into alignment with those around them.

The present research also compliments and extends the literature on the effect of mental simulation on self-knowledge. It has been demonstrated that thinking about other people can make participants think of themselves as being more similar to the simulated other (52, 56, 66). Thus, it is possible that the conversation partners in our studies who achieved better predictions of each other did so via mental simulation. More accurate simulation in conversations with greater inter-self alignment would also be consistent with work demonstrating that mental simulation can bolster feelings of social connection (100). Future research is needed to help clarify the link between inter-self alignment, mental simulation, and enjoyment.

Inter-self alignment is similar to, although still distinct from, two other social psychological constructs. The "chameleon effect" refers to the automatic tendency to mimic the expressions and postures of others to promote bonding (101, 102). While this phenomenon also emphasizes the emergence of self-other similarity, it focuses on behavioral expressions of similarity (e.g., nodding), rather than people's internal views of themselves. Inter-self alignment is also similar to the construct of "informational conformity," in which people internalize other people's beliefs about the world. To our knowledge, however, research on informational conformity tends to make the relevant beliefs salient (31), for instance, with explicit feedback from an interaction partner about their opinion (103). In contrast, inter-self alignment emerges even though we did not ask participants to compare themselves to their partner and occurred equivalently for shallow and deep conversations—despite the latter being more likely to activate self-knowledge. To our knowledge, inter-self alignment characterizes a previously unexplored phenomenon of self-views spontaneously aligning through conversation.

Some limitations constrain the generalizability and scope of the conclusions which can be drawn from the reported findings. First, the relationship we observed between inter-self alignment and conversation enjoyment was not causal. Based on our preregistered hypotheses, we describe inter-self alignment as a predictor of enjoyment. With our current data, we cannot know, however, whether inter-self alignment leads to greater conversational enjoyment, whether the inverse is true, or whether a third variable predicts both. Second, the participants in our study were recruited from an undergraduate population which limits the generalizability of our findings. Undergraduate students are younger than the general population and may have a less firmly established sense of self, potentially making them more willing to shift their self-views. Third, although we varied context in terms of conversational depth and whether conversations took place in person vs. online, there are many other contexts that would be important to test in future studies (e.g., between close others, conversations of different lengths, or repeated conversations). Our particular sample consisted of students in a relatively novel social network who may have been more eager to connect with their peers compared to others in more established social networks.

In summary, we demonstrate that conversations align self-views and that the more conversation partners' self-views converge, the more they enjoy their conversations. These effects were robust to whether the conversation was about shallow or deep topics. We also found that conversation tends to make people see themselves in more normative ways, by adjusting their idiosyncratic self-views. Future research is needed to understand whether and how these effects may compound with repeated interactions—i.e., to what extent we become the company we keep. We are excited by recent methodological and analytical advances that make these and other questions about social interaction newly tractable (104–107).

Our social understanding of the self parallels the social understanding of language. Just as Wittgenstein noted that "the meaning of a word is its use in the language"(108), we find that our view of the self is dynamically forged in social interaction. Through the back-and-forth of dialogue, our self-views mutually evolve. The most enjoyable conversations are those in which we come to see ourselves in the same way.

Data, Materials, and Software Availability. Survey materials and anonymized csv files with preprocessed data for all analyses in the paper have been deposited in OSF (https://osf.io/7wtz2/?view_only=8ee86d31444649d9a9a9ed218b932883) (109).

1. P. T. Costa, R. R. McCrae, Personality stability and its implications for clinical psychology. *Clin. Psychol. Rev.* **6**, 407–423 (1986).
2. M. Diehl, L. M. Jacobs, C. T. Hastings, Temporal stability and authenticity of self-representations in adulthood. *J. Adult Dev.* **13**, 10–22 (2006).
3. W. B. Swann Jr. "Self-verification theory" in *Handb. Theor. Soc. Psychol.* E. Higgins, Ed. (2012), vol. 2, pp. 23–42.
4. W. B. Swann, C. A. Hill, When our identities are mistaken: Reaffirming self-conceptions through social interaction. *J. Pers. Soc. Psychol.* **43**, 59–66 (1982).
5. V. Burr, *An Introduction to Social Constructionism* (Routledge, 1995).
6. K. J. Gergen, *The Social Construction of Self* (Oxford University Press, 2011).
7. S. E. Hampson, The Construction of Personality: An Introduction: (Taylor & Francis Routledge, 1988), vol. 2.
8. H. Markus, E. Wurf, The dynamic self-concept: A social psychological perspective. *Annu. Rev. Psychol.* **38**, 299–337 (1987).
9. S. Sinclair, J. Huntsinger, J. Skorinko, C. D. Hardin, Social tuning of the self: Consequences for the self-evaluations of stereotype targets. *J. Pers. Soc. Psychol.* **89**, 160–175 (2005).
10. M. Snyder, "When belief creates reality" in *Adv. Exp. Soc. Psychol.* L. Berkowitz, Ed. (1984), vol. 18, pp. 247–305.
11. R. H. Fazio, E. A. Effrein, V. J. Falender, Self-perceptions following social interaction. *J. Pers. Soc. Psychol.* **41**, 232–242 (1981).
12. R. Dunbar, *Grooming, Gossip, and the Evolution of Language* (Harvard University Press, 1996).
13. R. I. M. Dunbar, Gossip in evolutionary perspective. *Rev. Gen. Psychol.* **8**, 100–110 (2004).
14. A. M. Mastroianni, D. T. Gilbert, Do conversations end when people want them to? *Proc. Natl. Acad. Sci. USA* **118**, e2011809118 (2021), https://www.pnas.org/doi/abs/10.1073/pnas.2011809118.
15. S. E. Brennan, H. H. Clark, Conceptual pacts and lexical choice in conversation. *J. Exp. Psychol. Learn. Mem. Cogn.* **22**, 1482–1493 (1996).
16. R. D. Hawkins, M. C. Frank, N. D. Goodman, Characterizing the dynamics of learning in repeated reference games. *Cogn. Sci.* **44**, e12845. (2020).
17. A. Stolk, L. Verhagen, I. Toni, Conceptual alignment: How brains achieve mutual understanding. *Trends Cogn. Sci.* **20**, 180–191 (2016).
18. C. J. Wynn, T. S. Barrett, V. Berisha, J. M. Liss, S. A. Borrie, Speech entrainment in adolescent conversations: A developmental perspective. *J. Speech Lang. Hear. Res.* **66**, 3132–3150 (2023).
19. H. H. Clark, S. E. Brennan, "Grounding in communication" in *Perspectives on Socially Shared Cognition*, L. B. Resnick, J. M. Levine, S. D. Teasley, Eds. (American Psychological Association, 1991), pp. 127–149.
20. A. Coman, I. Momennejad, R. D. Drach, A. Geana, Mnemonic convergence in social networks: The emergent properties of cognition at a collective level. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 8171–8176 (2016).
21. M. J. Pickering, S. Garrod, Alignment as the basis for successful communication. *Res. Lang. Comput.* **4**, 203–228 (2006).
22. B. Sievers, C. Welker, U. Hasson, A. M. Kleinbaum, T. Wheatley, Consensus-building conversation leads to neural alignment. *Nat. Commun.* **15**, 3936 (2024).
23. G. Kossinets, D. J. Watts, Origins of homophily in an evolving social network. *Am. J. Sociol.* 115, 405–450 (2009).
24. B. Laursen, Making and keeping friends: The importance of being similar. *Child Dev. Perspect.* **11**, 282–289 (2017).
25. M. McPherson, L. Smith-Lovin, J. M. Cook, Birds of a feather: Homophily in social networks. *Annu. Rev. Sociol.* **27**, 415–444 (2001).
26. C. Parkinson, A. M. Kleinbaum, T. Wheatley, Similar neural responses predict friendship. *Nat. Commun.* **9**, 332 (2018).
27. L. M. Verbrugge, The structure of adult friendship choices. *Soc. Forces* **56**, 576–597 (1977).
28. S. Aral, L. Muchnik, A. Sundararajan, Distinguishing influence-based contagion from homophily-driven diffusion in dynamic networks. *Proc. Natl. Acad. Sci. U.S.A.* **106**, 21544–21549 (2009).
29. L. Ma, R. Krishnan, A. L. Montgomery, Latent homophily or social influence? An empirical analysis of purchase within a social network *Manag. Sci.* **61**, 454–473 (2015).
30. M. H. W. van Zalk, S. Nestler, K. Geukes, R. Hutteman, M. D. Back, The codevelopment of extraversion and friendships: Bonding and behavioral interaction mechanisms in friendship networks. *J. Pers. Soc. Psychol.* **118**, 1269–1290 (2020).
31. R. B. Cialdini, N. J. Goldstein, Social influence: Compliance and conformity. *Annu. Rev. Psychol.* **55**, 591–621 (2004).
32. M. Deutsch, H. B. Gerard, A study of normative and informational social influences upon individual judgement. *J. Abnorm. Psychol.* **51**, 629–636 (1955).
33. D. Dolinski, M. Nawrat, I. Rudak, Dialogue involvement as a social influence technique. *Pers. Soc. Psychol. Bull.* **27**, 1395–1406 (2001).
34. M. Sherif, *The Psychology of Social Norms* (Harper, 1936).
35. M. Vlasceanu, M. J. Morais, A. Duker, A. Coman, The synchronization of collective beliefs: From dyadic interactions to network convergence. *J. Exp. Psychol. Appl.* **26**, 453–464 (2020).
36. M. D. Back, S. C. Schmukle, B. Egloff, Becoming friends by chance. *Psychol. Sci.* 19, 439–440 (2008).
37. D. Byrne, J. A. Buehler, A note on the influence propinquity upon acquaintanceships. *J. Abnorm. Psychol.* **51**, 147–148 (1955).
38. L. Festinger, S. Schachter, K. Back, *Social Pressures in Informal Groups; A Study of Human Factors in Housing*. (Harper, 1950).
39. J. M. Rohrer, T. Keller, F. Elwert, Proximity can induce diverse friendships: A large randomized classroom experiment. *PLoS One* **16**, e0255097 (2021).
40. G. Echterhoff, E. T. Higgins, J. M. Levine, Shared reality: Experiencing commonality with others' inner states about the world. *Perspect. Psychol. Sci.* **4**, 496–521 (2009).
41. G. Echterhoff, E. T. Higgins, Shared reality: Construct and mechanisms. *Curr. Opin. Psychol.* **23**, iv–vii (2018).
42. M. Rossignac-Milon, N. Bolger, K. S. Zee, E. J. Boothby, E. T. Higgins, Merged minds: Generalized shared reality in dyadic relationships. *J. Pers. Soc. Psychol.* **120**, 882–911 (2020), 10.1037/pspi0000266.
43. U. Wagner *et al.*, Beautiful friendship: Social sharing of emotions improves subjective feelings and activates the neural reward circuitry. *Soc. Cogn. Affect. Neurosci.* **10**, 801–808 (2015).
44. A. Reece *et al.*, The CANDOR corpus: Insights from a large multimodal dataset of naturalistic conversation. *Sci. Adv.* **9**, eadf3197 (2023).
45. G. Cooney, T. Wheatley "On conversation" in *Handbook of Social Psychology*, D. T. Gilbert, S. Fiske, E. Finkel, W. Mendes, Eds. (n.d.).
46. C. Bassetti, K. Liberman, Making talk together: Simultaneity and rhythm in mundane Italian conversation. *Lang. Commun.* **80**, 95–113 (2021).
47. G. Cooney, A. M. Mastroianni, N. Abi-Esber, A. W. Brooks, The many minds problem: Disclosure in dyadic versus group conversation. *Curr. Opin. Psychol.* **31**, 22–27 (2020).
48. M. Yeomans, M. E. Schweitzer, A. W. Brooks, The conversational circumplex: Identifying, prioritizing, and pursuing informational and relational motives in conversation. *Curr. Opin. Psychol.* **44**, 293–302 (2022).
49. M. R. Mehl, S. Vazire, S. E. Holleran, C. S. Clark, Eavesdropping on happiness: Well-being is related to having less small talk and more substantive conversations. *Psychol. Sci.* **21**, 539–541 (2010).
50. S. Sprecher, Closeness and other affiliative outcomes generated from the Fast Friends procedure: A comparison with a small-talk task and unstructured self-disclosure and the moderating role of mode of communication. *J. Soc. Pers. Relat.* **38**, 1452–1471 (2021).
51. J. Sun, K. Harris, S. Vazire, Is well-being associated with the quantity and quality of social interactions? *J. Pers. Soc. Psychol.* **119**, 1478–1496 (2020).
52. M. J. Schneider, J. Rubin-McGregor, D. Tamir Simulation requires activation of self-knowledge to change self-concept. PsyArXiv [Preprint] (2022). https://psyarxiv.com/92mru/download (Accessed 7 March 2023).
53. P. Green, C. J. MacLeod, SIMR : an R package for power analysis of generalized linear mixed models by simulation. *Methods Ecol. Evol.* **7**, 493–498 (2016).
54. E. M. Templeton, L. J. Chang, E. A. Reynolds, M. D. C. LeBeaumont, T. Wheatley, Fast response times signal social connection in conversation. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2116915119 (2022).
55. M. Kardas, A. Kumar, N. Epley, Overly shallow?: Miscalibrated expectations create a barrier to deeper conversation *J. Pers. Soc. Psychol.* **122**, 367–398 (2022).
56. M. L. Meyer, Z. Zhao, D. I. Tamir, Simulating other people changes the self. *J. Exp. Psychol. Gen.* **148**, 1898–1913 (2019).
57. M. Snyder, Self-monitoring of expressive behavior. *J. Pers. Soc. Psychol.* **30**, 526 (1974).
58. N. E. Adler, E. S. Epel, G. Castellazzo, J. R. Ickovics, Relationship of subjective and objective social status with psychological and physiological functioning: Preliminary data in healthy white women. *Health Psychol.* **19**, 586–592 (2000).
59. D. Russell, L. A. Peplau, C. E. Cutrona, The revised UCLA Loneliness Scale: Concurrent and discriminant validity evidence. *J. Pers. Soc. Psychol.* **39**, 472–480 (1980).
60. H. A. Elfenbein *et al.* The choice that chooses you back: Introducing social variety-seeking as a trait predicting greater network brokerage through building connections with previously disconnected others. *SSRN* (2023).
61. Q. Tan *et al.*, Closer the relatives are, more intimate and similar we are: Kinship effects on self-other overlap. *Pers. Individ. Dif.* **73**, 7–11 (2015).
62. A. Aron, E. N. Aron, M. Tudor, G. Nelson, Close relationships as including other in the self. *J. Pers. Soc. Psychol.* **60**, 241–253 (1991).
63. D. Bates, M. Mächler, B. Bolker, S. Walker, Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* **67**, 1–48 (2015).
64. A. Kuznetsova, P. B. Brockhoff, R. H. B. Christensen, lmerTest package: Tests in linear mixed effects models. *J. Stat. Softw.* **82**, 1–26 (2017).
65. J. E. Dumas, M. Johnson, A. M. Lynch, Likableness, familiarity, and frequency of 844 person-descriptive words. *Pers. Individ. Differ.* **32**, 523–531 (2002).
66. J. Rubin-McGregor, Z. Zhao, D. I. Tamir, Simulation induces durable, extensive changes to self-knowledge. *J. Exp. Soc. Psychol.* **98**, 104229 (2022).
67. R. B. Rutledge, N. Skandali, P. Dayan, R. J. Dolan, A computational and neural model of momentary subjective well-being. *Proc. Natl. Acad. Sci. U.S.A.* **111**, 12252–12257 (2014).
68. J. A. Hall, S. A. Andrzejewski, J. E. Yopchick, Psychosocial correlates of interpersonal sensitivity: A meta-analysis. *J. Nonverbal Behav.* **33**, 149–180 (2009).
69. M. Schmid Mast, J. A. Hall, The impact of interpersonal accuracy on behavioral outcomes. *Curr. Dir. Psychol. Sci.* **27**, 309–314 (2018).
70. D. Tingley, T. Yamamoto, K. Hirose, L. Keele, K. Imai, mediation: R Package for Causal Mediation Analysis. *J. Stat. Softw.* **59**, 1–38 (2014).
71. C. J. Honey, A. Mahabal, B. Bellana, Psychological momentum. *Curr. Dir. Psychol. Sci.* **32**, 284–292 (2023).
72. P. Kuppens, N. B. Allen, L. B. Sheeber, Emotional inertia and psychological maladjustment. *Psychol. Sci.* **21**, 984–991 (2010).

73. M. Sudhof, A. Goméz Emilsson, A. L. Maas, C. Potts, “Sentiment expression conditioned by affective transitions and social forces” in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.* (2014), pp. 1136–1145.
74. M. A. Thornton, D. I. Tamir, Mental models accurately predict emotion transitions. *Proc. Natl. Acad. Sci. U.S.A.* **114**, 5982–5987 (2017).
75. D. I. Tamir, M. A. Thornton, J. M. Contreras, J. P. Mitchell, Neural evidence that three dimensions organize mental state representation: Rationality, social impact, and valence. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 194–199 (2016).
76. D. I. Tamir, M. A. Thornton, Modeling the predictive social mind. *Trends Cogn. Sci.* **22**, 201–212 (2018).
77. M. A. Thornton, S. Wolf, B. J. Reilly, E. G. Slingerland, D. I. Tamir, The 3d mind model characterizes how people understand mental states across modern and historical cultures. *Affect. Sci.* **3**, 93–104 (2022).
78. M. A. Thornton, D. I. Tamir, People represent mental states in terms of rationality, social impact, and valence: Validating the 3d mind model. *Cortex* **125**, 44–59 (2020).
79. M. A. Thornton, Constructing a 3-dimensional affect dictionary with word vector embeddings. *Mark Allen Thornton: Blog*, 13 August 2018.). http://markallenthornton.com/blog/3daffect/ (Accessed 28 April 2023).
80. S. Holm, A simple sequentially rejective multiple test procedure. *Scand. J. Stat.* 6, 65–70 (1979).
81. E. C. Baek, E. B. Falk, Persuasion and influence: What makes a successful persuader? *Curr. Opin. Psychol.* **24**, 53–57 (2018).
82. J. R. Andrews-Hanna, J. S. Reidler, C. Huang, R. L. Buckner, Evidence for the default network's role in spontaneous cognition. *J. Neurophysiol.* **104**, 322–335 (2010).
83. M. L. Meyer, M. D. Lieberman, Why people are always thinking about themselves: Medial prefrontal cortex activity during rest primes self-referential processing. *J. Cogn. Neurosci.* **30**, 714–721 (2018).
84. F. J. M. Ruby, J. Smallwood, H. Engen, T. Singer, How self-generated thought shapes mood–the relation between mind-wandering and mood depends on the socio-temporal content of thoughts. *PLoS One* 8, e77554 (2013).
85. X. Song, X. Wang, Mind wandering in Chinese daily lives–an experience sampling study. *PLoS One* **7**, e44423 (2012).
86. W. James, *The Principles of Psychology* (Henry Holt and Co., 1890), vol. 697.
87. A. D. Galinsky, G. Ku, C. S. Wang, Perspective-taking and self-other overlap: Fostering social bonds and facilitating social coordination. *Group Process. Intergroup Relat.* 8, 109–124 (2005).
88. C. E. Waugh, B. L. Fredrickson, Nice to know you: Positive emotions, self-other overlap, and complex understanding in the formation of a new relationship. *J. Posit. Psychol.* **1**, 93–106 (2006).
89. M. J. Bernstein *et al.*, Social interactions in daily life: Within-person associations between momentary social experiences and psychological and physical health indicators. *J. Soc. Pers. Relat.* **35**, 372–394 (2018).
90. N. Lathia, G. M. Sandstrom, C. Mascolo, P. J. Rentfrow, Happier people live more active lives: Using smartphones to link happiness and physical activity. *PLoS One* **12**, e0160589 (2017).
91. M. J. Zawadzki, J. M. Smyth, H. J. Costigan, Real-time associations between engaging in leisure and daily health and well-being. *Ann. Behav. Med.* **49**, 605–615 (2015).
92. F. M. Deutsch, L. Sullivan, C. Sage, N. Basile, The relations among talking, liking, and similarity between friends. *Pers. Soc. Psychol. Bull.* **17**, 406–411 (1991).
93. M. Laakasuo *et al.*, Homophily in personality enhances group success among real-life friends. *Front. Psychol.* **11**, 710 (2020), 10.3389/fpsyg.2020.00710.
94. A. Asikainen *et al.*, Cumulative effects of triadic closure and homophily in social networks. *Sci. Adv.* **6**, eaax7310 (2020).
95. E. C. Baek *et al.*, Lonely individuals process the world in idiosyncratic ways. *Psychol. Sci.* **34**, 683–695 (2023).
96. E. C. Baek *et al.*, In-degree centrality in a social network is linked to coordinated neural activity. *Nat. Commun.* **13**, 1118 (2022).
97. T. W. Broom, S. Iyer, A. L. Courtney, M. L. Meyer, Loneliness corresponds with neural representations and language use that deviate from shared cultural perceptions. *Commun. Psychol.* **2**, 40 (2024).
98. A. L. Courtney, M. L. Meyer, Self-other representation in the social brain reflects social connection. *J. Neurosci.* **40**, 5616–5627 (2020).
99. U. Hasson *et al.*, Enhanced intersubject correlations during movie viewing correlate with successful episodic encoding. *Neuron* **57**, 452–462 (2008).
100. E. Baek, D. Tamir, E. Falk, Considering others' mental states causally increases feelings of social bonding and information sharing. PsyArXiv [Preprint] (2019). https://psyarxiv.com/nw43x/download?format=pdf (Accessed 20 March 2023).
101. T. L. Chartrand, J. A. Bargh, The chameleon effect: The perception-behavior link and social interaction. *J. Pers. Soc. Psychol.* **76**, 893–910 (1999).
102. J. L. Lakin, V. E. Jefferis, C. M. Cheng, T. L. Chartrand, The chameleon effect as social glue: Evidence for the evolutionary significance of nonconscious mimicry. *J. Nonverbal Behav.* **27**, 145–163 (2003).
103. J. Zaki, J. Schirmer, J. P. Mitchell, Social influence modulates the neural computation of value. *Psychol. Sci.* **22**, 894–900 (2011).
104. M. Dingemanse *et al.*, Beyond single-mindedness: A figure-ground reversal for the cognitive sciences. *Cogn. Sci.* **47**, e13230 (2023).
105. E. Redcay, L. Schilbach, Using second-person neuroscience to elucidate the mechanisms of social interaction. *Nat. Rev. Neurosci.* **20**, 495–505 (2019).
106. T. Wheatley, A. Boncz, I. Toni, A. Stolk, Beyond the isolated brain: The promise and challenge of interacting minds. *Neuron* **103**, 186–188 (2019).
107. T. Wheatley, M. A. Thornton, A. Stolk, L. J. Chang, The emerging science of interacting minds. *Perspect. Psychol. Sci.* **19**, 355–373 (2024).
108. L. Wittgenstein, *Philosophical Investigations* (Wiley-Blackwell, New York, NY, USA, 1953).
109. C. Welker, T. Wheatley, G. Cason, C. Gorman, M. Meyer, Selves in conversation. OSF. https://osf.io/7wtz2/?view_only=8ee86d31444649d9a9a9ed218b932883. Deposited 5 December 2023.